Field accuracy is not heat accuracy: wall-flux-constrained surrogates for steady two-dimensional non-Boussinesq convection

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Abstract

Surrogates for non-Boussinesq convection are ranked by relative L_2 field error. For steady two-dimensional enclosure convection, this metric can rank a model above doing nothing on wall heat transfer. On 200 held-out classical-cavity samples, a surrogate with 0.09% median field error misses the both-wall integral Nusselt number by 2.0% on average; in 43% of samples field error is below 1% while Nu error exceeds 2%. A 22-sample random-wall replication gives 0.47% median field error and 4.04% median Nu error. Against the same 64^2 reference, a wall-flux training penalty reduces the matched six-point errors to 0.01% and 0.58%. The target correction itself changes by 0.43–0.60 percentage points under grid refinement, so these values are model-relative diagnostics, not converged absolute heat-transfer accuracy. At ten points, the two six-point-trained wall-flux U-Nets span 1.00–2.21% versus 1.11% for returning X. A complete two-seed 4×4 training/evaluation matrix puts every row minimum on its own training stencil. A stencil-free control is worse at every stencil. The mechanism is a redistribution of the wall-normal gradient: at $\varepsilon = 1.2$ the hot-wall integral Nu changes by 0.67% while the local profile changes by 4.25% on the same grid. The correction factorises as $D = \varepsilon P(X) + O(\varepsilon^2)$ across 627 paired solutions with $\varepsilon = 0.02$–1.9: nominal velocity exponents are $m = 1.023 \pm 0.018$ and 0.995 ± 0.019, and A is a functional of the flow. A 134-wall-family cluster bootstrap on the 600-point core gives wider intervals and does not resolve the curvature magnitude: positive curvature is established for velocity and pressure, but not for temperature. The scaling makes public single-contrast data reusable within the tested boundary family: a plain model trained at $\varepsilon_0 = 1.2$ and rescaled by $\varepsilon/\varepsilon_0$ matches the factorized target to 0.2% relative. Comparisons in one alternate geometry, one alternate Prandtl number, two public families and one three-dimensional resolution are consistency checks, not evidence of universal transfer. At $\varepsilon = 1.9$ the integral Nu change is -5.1% at $Ra = 10^4$ and -1.5 to -1.8% at $Ra = 10^5$–10^6. Inference takes 6.3 ms; replacing the compressible member saves

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2.9× per pair.

Keywords: natural convection, Boussinesq approximation, low-Mach compressible flow, physics-informed surrogate, data reuse, wall heat flux

Nomenclature

Roman		*Greek*	
A	prefactor of $R = A\varepsilon^m$	ΔT	wall-to-wall temperature contrast
D	correction $Y - X$	ε	relative contrast, $\Delta T/T_0$
Fr	Froude number, $Fr^2 = \Delta T/T_0$	θ	normalised temperature
m	scaling exponent	κ	curvature in $R = A\varepsilon(1 + \kappa\varepsilon)$
Ma	Mach number	ρ	density
Nu	Nusselt number	χ	base-flow advection strength
p	pressure		
E, e	total and internal energy	$\boldsymbol{P}_{\mathrm{work}}$	pressure-work tensor
P, Q	first- and second-order correction fields		
Pr	Prandtl number		
R	relative correction, $\|Y - X\|/\|Y\|$		
Ra	Rayleigh number		
Re	Reynolds number, $(Ra/Pr)^{1/2}$		
T	temperature, T_0 its reference value		
$\boldsymbol{u}$	velocity, (u_x, u_y)		
$\hat{u}$	peak Boussinesq speed		
X, Y	Boussinesq and compressible solutions	$\mathcal{L}$	linearised low-Mach operator at $O(\varepsilon)$

1. Introduction

Steady natural convection is a workhorse of thermal engineering and it is expensive. A converged pair of two-dimensional Boussinesq and compressible solves on a 64^2 mesh costs 34 s of wall clock on an eight-way desktop (Sec. 4.5); three-dimensional problems cost hours to days. Here the scaling law and the speed-up are measured in two dimensions; the three-dimensional pairs are used only to check the exponent. Surrogate modelling removes this cost by learning the map from a cheap model to an expensive one [1, 2]. For convection the two models are not arbitrary fidelities: they are the Oberbeck–Boussinesq equations and the low-Mach compressible equations. Boussinesq is exact only as the relative temperature contrast vanishes; the difference is $O(\beta\Delta T)$ and has been characterised asymptotically for decades [3, 4].

The numerical literature offers several ways to reduce the cost. Reduced-order models replace the discretised equations by a small number of modes [5]; graph convolutional networks exploit the mesh [6, 7]; operator learning removes the mesh-resolution constraint [8, 9, 10], with applications to sparse-field reconstruction [11] and heat-pipe startup [12]. Physics-informed neural networks take the opposite route and solve one instance by minimising a residual [13, 14, 15, 16]; reviews of the numerical analysis and the application landscape are given by De Ryck and Mishra [17] and Luo et al. [18]. These methods reduce the cost of a single query, but they still need paired data at the operating conditions of interest.

Non-Oberbeck–Boussinesq effects have been measured under strong temperature contrast [19], in cryogenic helium [20], and in water-filled cavities [21]; at large contrast they can grow cold-corner vortices and reverse the large-scale circulation [22]. The heat-transfer and large-scale dynamics of Rayleigh–Bénard convection are reviewed by Ahlers et al. [23]. Low-Mach formulations for enclosed thermal convection have been developed and compared with the Boussinesq limit [24, 25], and the rigorous derivation of the Oberbeck–Boussinesq system has been revisited [26]. These studies fix the contrast and compare the two solutions. They do not extract its dependence as a reusable scaling.

The practical difficulty appears when a surrogate is trained on a paired data set at a single contrast. Menglik et al. [27] released 4000 paired Boussinesq→compressible steady solutions in two dimensions, with three held-out splits, at one value of $\varepsilon = \Delta T/T_0 = 1.2$. A network trained on that release learns the correction at $\varepsilon = 1.2$. At another contrast the Boussinesq input is unchanged, because the Boussinesq equations do not contain ε, while the correction that must be added scales with ε. The network has no signal with which to rescale what it memorised. The usual remedy is a second paired data set, which pays again the cost the surrogate was meant to remove.

This paper characterises that barrier and extracts a portable scaling rule. The correction between the two members factorises. An ε-expansion of the compressible member gives the Boussinesq system at $O(1)$ and a linear, ε-independent operator at $O(\varepsilon)$. Hence

$$D = Y - X = \varepsilon P(X) + \varepsilon^2 Q(X) + \cdots , \tag{1}$$

so a surrogate only has to learn the spatial shape P, and ε can be applied outside the network. The same expansion identifies the parameter that controls the gap: $Fr^2 = \varepsilon$ identically in the non-dimensionalisation used by the released data and by our solver, while Ma enters the retained equations only at $O(Ma^2)$. The break-symmetry axis is therefore one-dimensional.

We test the consequence on 627 paired solutions in the unit square and use a second geometry, a second Prandtl number, 33 three-dimensional pairs, and two public data sets as consistency checks. The law describes the velocity and pressure channels; the temperature channel has a flow-dependent amplitude that collapses on the peak Boussinesq speed. The scaling law makes a single-contrast public data set reusable within its boundary-condition family, and the limit is equally clear: the law transfers the ε dependence, not the flow-dependent prefactor.

The central result, however, is not the law by itself. The correction is largely a redistribution of wall flux: at $\varepsilon = 1.2$ the integral Nusselt number of the classical cavity changes by 0.67% on the hot wall while the local wall flux changes by 4.25% in L_1. A surrogate can therefore reproduce the temperature field to 0.47% in relative L_2 and still miss the integral Nu by 4%. On the 22 held-out random-wall samples at $\varepsilon = 1.1$–1.3, 66% of the field-only predictions on average across five seeds have a field error below 1% and a Nu error above 2%; on the 200 cavity samples, 43% on average across three seeds do. Relative L_2 is not a sufficient selection metric

for a heat-transfer surrogate. A wall-flux penalty in the training loss reduces the heat-transfer bias at a small cost in field error.

The contributions are:

1. a per-sample demonstration that relative L_2 field accuracy does not imply Nusselt-number accuracy, with a mechanism: the correction redistributes the wall-normal temperature gradient while the integral almost cancels;
2. a wall-flux training penalty that reduces the both-wall error against the 64^2 target from 4.0% to 0.58% on the random-wall replication (0.01% on the classical cavity) under the matched six-point estimator. A full two-seed 4×4 train/evaluate matrix shows that each training stencil is optimal only on its matching evaluation stencil; we make no claim of absolute or stencil-independent wall-flux accuracy;
3. a quantitative law for the Boussinesq–compressible gap, $R = A\varepsilon^m$ with m close to one over the full window and a positive curvature for velocity and pressure whose magnitude is not resolved, calibrated over two decades of ε with 134-family cluster uncertainty. Alternate geometry, fluid, public-family and 33-pair three-dimensional checks support consistency within their tested ranges; they are not a universality claim. The factorized target is one implementation; applying the known ε scaling is what makes the data reusable, so the factorisation, not the network, is the reusable object.

The main sweep is two-dimensional and steady. The three-dimensional evidence consists of 33 pairs at three Rayleigh numbers for the exponent and a 12-train/3-test leave-one-contrast ROM at five contrasts. It does not establish a three-dimensional wall-flux model or design workflow. The scope is stated in Sec. 5.

2. Physical scaling

2.1. Two members and one contrast parameter

Each sample is a paired steady solution of one configuration. The high-fidelity member Y is the low-Mach compressible formulation in $(\boldsymbol{u}, T, \rho, p)$:

$$\nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{2}$$

$$\nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u}) = -\nabla p + \frac{1}{Re} \nabla \cdot \boldsymbol{\sigma} - \frac{1}{Fr^2} \rho \boldsymbol{g}, \tag{3}$$

$$\nabla \cdot (\rho E \boldsymbol{u}) = \frac{\gamma}{Re\, Pr} \nabla^2 e + \nabla \cdot (\boldsymbol{P}_{\text{work}} \boldsymbol{u}) + O(Ma^2), \tag{4}$$

$$\rho e = \gamma Ma^2 p + 1, \qquad Re = (Ra/Pr)^{1/2}. \tag{5}$$

The low-fidelity member X is the Boussinesq system in $(\boldsymbol{u}, T, p)$ on the same mesh:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{6}$$

$$\boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \left(\frac{Pr}{Ra}\right)^{1/2} \nabla^2 \boldsymbol{u} + \theta \boldsymbol{g}, \tag{7}$$

$$\boldsymbol{u} \cdot \nabla \theta = (Ra\, Pr)^{-1/2} \nabla^2 \theta, \qquad \theta = \frac{T - T_0}{\Delta T}. \tag{8}$$

Both members use Taylor–Hood-type mixed elements, damped Newton iterations and a MUMPS factorisation. They start from the same initial field and the same wall-temperature seed, so they follow the same branch. Left and right walls are Dirichlet; top and bottom walls are Dirichlet when their Fourier dial is nonzero and adiabatic otherwise; velocity is no-slip on all walls. The local Nusselt profile is $-\partial\theta/\partial n$ from a six-point one-sided stencil, and the integral Nu is its wall average. The classical differentially heated cavity is the one-mode member of the same family.

The generator draws Ra and ε independently. At fixed Ra, changing ε changes $Fr = \sqrt{\varepsilon}$, the buoyancy scale, while Re, Pr and γ stay fixed. The Boussinesq member sees only Ra and the thermal boundary data. This is a statement about the dimensionless parameterisation; a fixed apparatus follows the one-dimensional slice $Ra \propto \varepsilon$, while the (Ra, ε) rectangle contains that slice.

Two simplifications follow from the equations. First, $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically in this non-dimensionalisation, so Fr is not a second free axis. Second, Ma appears only at $O(Ma^2)$ in the retained low-Mach system. A 24-fold change in Ma at fixed Ra, T_0 and ΔT leaves the measured correction unchanged to a relative 10^{-5}, which is the precision of the solve. The free parameter is ε.

The Boussinesq solution is ε-blind to solver precision. Neither T_0 nor ΔT appears alone in Eqs. (6)–(8); with the thermal data prescribed in θ and Ra fixed, X cannot depend on ε. Across 20 configurations $\times$ 5 contrasts, the maximum relative deviation of X is 9.77×10^{-8}, the solver's convergence floor. The amplitude of the correction is thus set by a variable that is absent from the input. The spatial shape remains a function of X and is learnable.

2.2. Why the correction factorises

Write $T = T_0(1 + \varepsilon\theta)$ with $\theta = O(1)$ and use $\rho = 1/T$. The buoyancy source becomes

$$\frac{\rho \boldsymbol{g}}{Fr^2} = \frac{\rho \boldsymbol{g}}{\varepsilon} = \frac{\boldsymbol{g}}{\varepsilon} - \theta \boldsymbol{g} + \varepsilon\theta^2 \boldsymbol{g} + \cdots . \tag{9}$$

The constant $\boldsymbol{g}/\varepsilon$ is balanced by a hydrostatic pressure p_{-1}/ε with $\nabla p_{-1} = -\boldsymbol{g}$, which drops out at the next order. The $O(1)$ balances are

$$\nabla \cdot \boldsymbol{u}_0 = 0, \tag{10}$$

$$\boldsymbol{u}_0 \cdot \nabla \boldsymbol{u}_0 = -\nabla \tilde{p}_0 + Re^{-1} \nabla^2 \boldsymbol{u}_0 + \theta_0 \boldsymbol{g}, \tag{11}$$

$$\boldsymbol{u}_0 \cdot \nabla \theta_0 = (Re\, Pr)^{-1} \nabla^2 \theta_0, \tag{12}$$

which is term-by-term the Boussinesq system solved as X, with $Re^{-1} = (Pr/Ra)^{1/2}$. The correction at $O(\varepsilon)$ is driven by the density modulation:

$$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0), \tag{13}$$

$$\nabla \cdot (\boldsymbol{u}_1 \otimes \boldsymbol{u}_0 + \boldsymbol{u}_0 \otimes \boldsymbol{u}_1 - \theta_0 \boldsymbol{u}_0 \otimes \boldsymbol{u}_0) = -\nabla \tilde{p}_1 + Re^{-1} \nabla^2 \boldsymbol{u}_1 + \theta_1 \boldsymbol{g} - \theta_0^2 \boldsymbol{g}. \tag{14}$$

The mass equation is inhomogeneous, so the operator is not the Boussinesq operator linearised about X; it is the linearisation of the retained low-Mach system about X, with the mass constraint carried inside. Its only singular mode is the constant pressure, and the compatibility condition holds because $\boldsymbol{u}_0 \cdot \boldsymbol{n} = 0$ on the walls. Since the operator is built from X and contains no explicit ε,

$$D = -\varepsilon \mathcal{L}^{-1} S_1[X] - \varepsilon^2 \mathcal{L}^{-1} S_2[X] + \cdots = \varepsilon P(X) + \varepsilon^2 Q(X) + \cdots . \tag{15}$$

Five predictions follow: $R \to 0$ as $\varepsilon \to 0$; $R \propto \varepsilon$ with a vanishing intercept; the shape of D does not rotate with ε; $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$; and the ε^1 term carries most of the correction energy. All five hold numerically (Supplementary Material). The ε^1 term carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, and three terms carry 99.9% over the sampled range. The divergence identity has cosine 0.994 for $\varepsilon < 0.6$ and 0.901 at $\varepsilon = 1.9$; the drop is the footprint of the ε^2 term.

The expansion is valid for $\varepsilon < 2$, because the cold wall $T_c = T_0 - \Delta T/2$ must remain positive. At $\varepsilon = 1.9$, $T_c/T_0 = 0.05$, so the density expansion is close to its limit. This is the quantitative boundary of the law, not a fitting choice.

2.3. The measured law

The calibration set contains 600 random-boundary samples spanning 134 wall-mode families, with $Ra \sim \log U[10^2, 10^6]$ and $\varepsilon \sim U[0.2, 1.9]$, plus 27 low-contrast checks at $\varepsilon = 0.02, 0.05, 0.10$ and $Ra = 10^3, 10^4, 10^5$, with three independent seeds per condition. The $600 + 27 = 627$ points enter the fit; the low-contrast points anchor the $\varepsilon \to 0$ end. The relative correction is $R = \|Y - X\|/\|Y\|$.

Table 1 and Fig. 1 give the result. For u_x and u_y the full 627-point fit returns nominal exponents $m = 1.023 \pm 0.018$ and 0.995 ± 0.019. The pressure channel gives $m = 1.009 \pm 0.015$ in the fixed gauge described below; it is a numerical observation, not a gauge-invariant exponent. The intercepts are consistent with zero. These $\pm$ values are conventional pointwise standard errors and are not treated as family-level uncertainties.

The 600-point random-wall core contains only 134 independent wall-mode families, with up to 34 samples in one family. Resampling whole families gives 95% exponent intervals 1.030–1.136 (u_x), 0.939–1.058 (u_y), 1.075–1.166 (p) and 0.985–1.451 (θ) on the 600-point core. The 27 low-contrast points are not members of these families and are excluded from the cluster resampling; they only anchor the full 627-point fit. The full-fit exponents should therefore be read as approximate leading-order values, not as precision measurements of unity.

A quadratic fit $R = a\varepsilon + c\varepsilon^2 + b$ gives point estimates $\kappa_{\text{fit}} = c/a = 0.70/0.83/0.42/0.73$ for $u_x/u_y/\theta/p$. The corresponding 134-family cluster intervals are 0.15–3.14, 0.14–5.76, −4.90–7.06 and 0.22–2.80. Thus the sign is resolved for velocity and pressure, but not for temperature, and the curvature magnitude is not. A compact description is $R = A\varepsilon[1 + O(\varepsilon)]$ rather than an exact power. Fitting only $\varepsilon \geq 0.8$ returns $m = 1.23$–1.32, the window average of the same curvature.

Table 1: Power-law fit $R = A\varepsilon^m$ over the full two-decade window (627 samples). The intercept is fitted separately on $\varepsilon \leq 0.5$. The pressure row uses the fixed gauge of Sec. 2.5; its exponent is not gauge-invariant. θ is not included: its free power-law fit is not identifiable, and its amplitude is discussed separately.

channel	A	m	R^2	intercept ($\varepsilon \leq 0.5$)
u_x	0.147	1.023 ± 0.018	0.84	$+0.0003 \pm 0.0047$
u_y	0.193	0.995 ± 0.019	0.82	$+0.0030 \pm 0.0062$
p	0.148	1.009 ± 0.015	0.87	$+0.0016 \pm 0.0030$

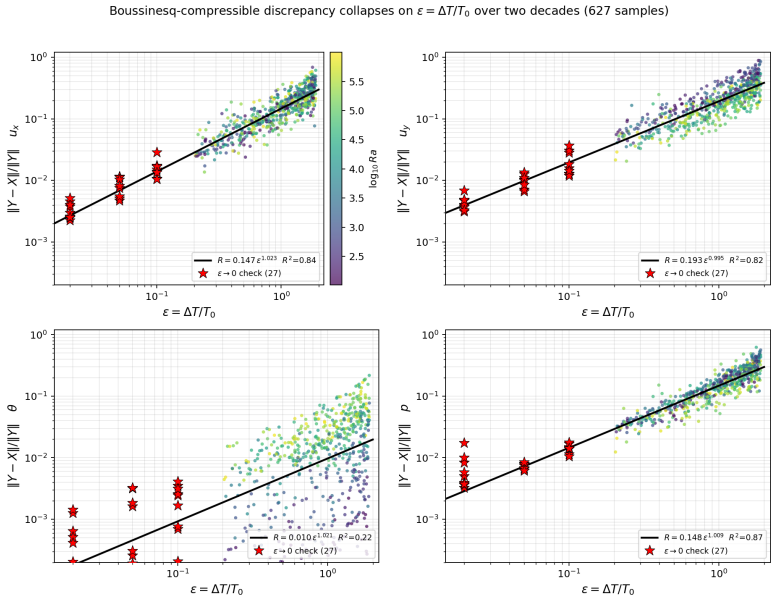


Figure 1: The Boussinesq–compressible discrepancy collapses on $\varepsilon = \Delta T/T_0$ for u_x, u_y and p over two decades. The θ panel scatters because its amplitude depends on the flow; it collapses on the peak Boussinesq speed (Supplementary Material). Points: 600 random-boundary samples. Stars: 27 low-contrast checks. Lines: power-law fits of Table 1 for u_x, u_y and p; the θ line is a free fit ($R^2 = 0.22$) and is not part of the common-exponent claim.

The temperature channel needs a flow-dependent amplitude. Its free power law has $R^2 = 0.22$ and a slope that moves from 1.02 over the full window to 1.67 on the 104-configuration subset; the fitted amplitude is not predictive. This is not a

failure of the expansion. Writing $R_\theta/\varepsilon = c\hat{u}^a$, where $\hat{u}$ is the peak Boussinesq speed available from X, collapses six data sets from two independent generators: $a = 1.86$–1.98 and $c = 0.54$–0.69 for random-wall families, $a = 2.36$ and $c = 2.06$–2.10 for the classical cavity. The classical cavity has a single strong circulation with $\hat{u}$ 41% above the random-wall median and 56% above ours. The same reading explains the residual scatter in u_x, u_y and p: the prefactor is a functional of the flow, and the flow is in the input X.

2.4. What changes and what does not

The exponent is stable; the prefactor is not. In a 2:1 cavity with 104 samples, the ε-only fit gives $m = 1.09/1.16/0.93/1.08$ for $u_x/u_y/\theta/p$, within 0.3–1.8 combined standard errors of the unit-square values. The prefactor moves from 0.147/0.193/0.010/0.148 to 0.175/0.175/0.008/0.222. Changing the fluid from $Pr = 0.71$ to $Pr = 4.38$ in 104 strictly paired configurations leaves the exponent within 1.5% and the paired median R within 5%; the prefactor moves by at most 3.3% except for pressure (6.1%). A 33-sample three-dimensional check at $Ra = 10^4$, 3×10^4 and 10^5 ($res = 16$) returns exponents for u_x, u_y and p of 1.09/1.05/1.14, 1.03/1.04/1.09 and 1.01/1.01/1.09, all within 0.15 of unity. The prefactor varies by factors of 2–13 across the three wall types.

The practical reading is simple. The law supplies the ε dependence. It does not supply the flow dependence. A surrogate must therefore take X as input rather than a table of A. This is also why transfer across boundary-condition families fails: the model has learned one family's prefactor, not a universal constant.

2.5. The pressure channel

The two members store pressure in different conventions. The compressible raw pressure carries a hydrostatic ramp 12 times larger than its dynamic part, and the two weak forms use opposite signs. Comparing raw pressures measures the conventions, not the flow. We subtract the mean, remove a least-squares linear fit in the gravity direction, and flip the sign of the Boussinesq member. Under this gauge the per-sample median of $|X_p|/|Y_p|$ is 0.960 and the cosine is +0.991, against 0.929 and +0.986 for the public training set. The pressure channel then follows the same law as the velocities, with $A = 0.148$ and $m = 1.009$. All pressure numbers use this fixed gauge; the exponent is not gauge-invariant.

3. Surrogate design

3.1. Residual learning and the factorised target

All surrogates use residual learning,

$$\hat{Y} = X + N(X), \tag{16}$$

with a U-Net of width 32 or an FNO with 12 modes, width 32 and four layers (1.19M parameters). Training uses Adam with cosine annealing, 200 epochs and batch 16. Inputs and targets are scaled per channel by training-set standard deviations.

The prior changes the target, not the backbone. Instead of learning D, the network learns $P = D/\varepsilon$ at the available contrast and the model returns

$$\hat{Y} = X + \varepsilon N(X). \tag{17}$$

We compare five variants:

- A: ε-blind, target D;
- B: ε supplied as a constant input channel, target D;
- P1: ε supplied as input, target D/ε, multiplied outside;
- P2: as P1 with a learned scalar curvature, $\varepsilon(1 + \kappa\varepsilon)$;
- P3: as P1 with no ε input, so the network sees only X.

P1 is the most accurate form when ε is known at training time. P3 is the form that can be trained on a single-contrast data set and applied at another contrast; it must infer the prefactor from the flow. The distinction is the difference between a model that interpolates in ε and one that reuses a data set.

3.2. Wall-flux loss

Field accuracy does not guarantee heat-transfer accuracy. For applications we add a relative L_1 penalty on the wall-normal gradient,

$$\mathcal{L} = \mathcal{L}_{\text{field}} + \lambda \frac{\sum_{\text{wall}} |\hat{q} - q|}{\sum_{\text{wall}} |q|}, \qquad q = -\frac{\partial \theta}{\partial n}, \tag{18}$$

with $\lambda = 1$ and a six-point one-sided stencil. The same models are evaluated with four-, six-, eight- and ten-point stencils to separate the loss from the estimator. The six-point result is a matched training/evaluation diagnostic and is not independent evidence. The four-, eight- and ten-point results are sensitivity checks. A separate stencil-free control below checks whether the result can be reproduced without wall-gradient training.

3.3. Data, splits and metrics

The main data set is the 600-sample random-boundary scan, with the 27 low-contrast checks used for the law. The paired scan contains 20 configurations $\times$ 5 contrasts; within a configuration only ε changes, so X is identical and the error of any ε-blind model is bounded exactly. The public release of Menglik et al. [27] provides 4000 random-wall pairs at $\varepsilon = 1.2$, a 200-pair in-distribution test split, a 200-pair out-of-distribution split with more complex wall dials, and 5000 + 200

classical cavity pairs. The 2:1 cavity and the $Pr = 4.38$ data are generated here with the same solver.

Two splits are used throughout: extrapolation (train $\varepsilon \leq 1.3$, test $\varepsilon > 1.3$, $n = 218$) and interpolation (stratified 2:1 in ε, $n = 200$). Errors are per-channel relative L_2 against Y; the floor is $\hat{Y} = X$. Training uses three seeds unless stated otherwise. When a model trained on one data set is applied to another, the target is scaled by the source per-channel standard deviation; re-normalising by the target's own standard deviation changes the temperature channel by 21% and is not used.

4. Results

4.1. Field accuracy is not heat accuracy

We first ask whether the metric used to select these surrogates is the right one. Table 2 and Fig. 2 compare, for the same 200 held-out classical-cavity samples at $\varepsilon = 1.2$, the per-sample relative L_2 error of the temperature field with the per-sample error of the integral Nusselt number. The models are the ones of Sec. 4.2: a U-Net trained on the 4000 public random-wall pairs at $\varepsilon = 1.2$ and applied without retraining to the classical-cavity split.

The field-only prior reduces the median field error from 8.4% (returning X) to 0.09%. Its median integral-Nu error is 1.4–2.9% across the three seeds, against 0.48% for returning X, and the fraction of samples with a field error below 1% and a Nu error above 2% is 43% on average (23–82% across seeds). Within a seed the two errors are anti-correlated (Spearman -0.73 for one seed; the plotted points are three-seed averages): the samples whose field looks best are not the samples whose heat transfer is best. The wall-flux penalty moves the same model back into the low-error region under the six-point stencil: median Nu error 0.01%, local wall-flux error below 0.1%, and field error 0.10%.

The pattern is not an artefact of crossing boundary-condition families. The full held-out random-wall split has 200 samples, of which 22 lie at $\varepsilon = 1.1$–1.3. On those 22, a prior trained on the other 400 random-wall samples has a 0.47% median temperature-field error and a 4.04% median integral-Nu error. The wall-flux penalty reduces the median Nu error to 0.58% and the median local flux error from 4.94% to 1.14%, while the median field error moves from 0.47% to 0.49%. On average across five seeds, 66% of these samples (50–77% across seeds) have the field error below 1% and the Nu error above 2%. The field norm is insensitive to the wall gradient in both families.

The mechanism is visible in the correction itself. The compressible member adds heat over the lower half of the cavity and removes it over the upper half; the integral Nu is the Nu-weighted mean of this redistribution and almost cancels, while the wall-normal gradient changes by several per cent. A field norm dominated by the interior does not see this boundary-layer bias. Section 4.3 gives the per-wall breakdown, the stencil dependence and the Rayleigh dependence.

Table 2: Field error versus wall heat transfer on 200 held-out classical-cavity samples at $\varepsilon = 1.2$. T is the per-sample median relative L_2 error; the Nu columns are the hot wall, cold wall, and both walls normalised by the true total; the last two columns are the mean absolute wall-flux profile error and the fraction of samples with $T < 1\%$ and $Nu > 2\%$. All errors are relative to the 64^2 target. Values are means across three seeds; for the field-only prior the both-wall error ranges from 1.4 to 2.9% and the fraction from 23 to 82%. The wall-flux row uses the penalty of Eq. (18).

model	T median	Nu hot	Nu cold	Nu both	local flux	fraction
return X	8.4%	0.67%	0.64%	0.48%	5.6%	0%
field-only prior	0.09%	2.0%	2.1%	2.0%	4.5%	43%
prior + wall-flux	0.10%	**0.01%**	**0.01%**	**0.01%**	**0.08%**	0%

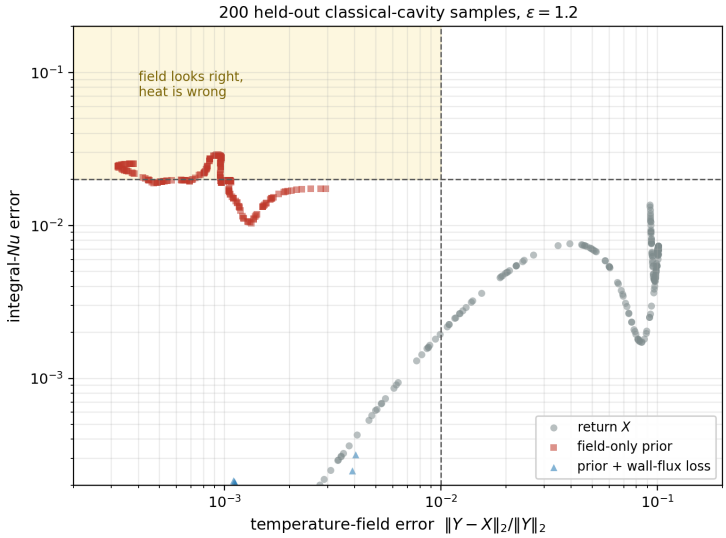


Figure 2: Per-sample temperature-field error against integral-Nu error for the 200 held-out classical-cavity samples at $\varepsilon = 1.2$. The shaded region is where a model looks accurate in the field but is wrong in heat transfer. The field-only prior sits there; the wall-flux penalty moves it out. Points are averages over three seeds. The prior's median Nu error is 1.4–2.9% across seeds; the 43% entry is the mean of the per-seed fraction, which spans 23–82%.

4.2. A single-contrast data set becomes reusable

Table 3 and Fig. 3 show the central result. A plain U-Net trained on the public 4000 random-wall pairs at $\varepsilon = 1.2$ reaches 0.142 in u_x on our extrapolation split, barely better than returning X (0.3018). Applying the known scaling to its output, $\hat{Y} = X + (\varepsilon/\varepsilon_0)N_{\text{plain}}(X)$, reaches 0.091. Training the same architecture on D/ε reaches 0.091, within a few per cent. The 1.55-fold gain over the unscaled plain model therefore comes from applying ε, not from the network parameterization. The prior's value is that it makes the scaling explicit and can be trained without ε as an input.

Channel-averaged, the error falls from 0.129 to 0.085, a factor 1.5. Relative to returning X, the channel average is 3.1× lower (3.3× in u_x). The public prior is better than the same prior trained on our own 382 samples (0.1250), because it sees ten times more configurations.

Table 3: Transfer from a single-contrast public data set to our $\varepsilon \in [0.2, 1.9]$ samples. Entries are relative L_2 errors for $u_x/u_y/\theta/p$. Both public sets are at $\varepsilon \equiv 1.2$. The "own data" row uses 382 training samples for extrapolation and 400 for interpolation, with the P3 prior (150 epochs, one seed). The "plain, rescaled" row applies $\hat{Y} = X + (\varepsilon/\varepsilon_0)N_{\text{plain}}(X)$ to the plain model's output; it is the fair baseline for the prior. The public 4000 rows use three seeds; the 5000 rows use one, so differences below 0.003 there are not resolved. The 0.2% quoted in the text is the difference between the "plain, rescaled" and "prior" rows (0.0912 vs 0.0914), not their absolute error.

training data	model	split	u_x	u_y	θ	p
do nothing	$\hat{Y} = X$	extrap.	0.3018	0.3731	0.0556	0.3320
		interp.	0.2203	0.2683	0.0381	0.2386
public 4000, random walls	plain	extrap.	0.1417	0.1717	0.0225	0.1811
		interp.	0.1195	0.1437	0.0189	0.1422
public 4000, random walls	plain, rescaled	extrap.	0.0912	0.1081	0.0118	0.1287
		interp.	0.0649	0.0766	0.0078	0.0871
public 4000, random walls	prior	extrap.	0.0914	0.1080	0.0118	0.1287
		interp.	0.0651	0.0767	0.0077	0.0872
own data, random walls	prior	extrap.	0.1250	0.1321	0.0170	0.1719
		interp.	0.0791	0.0811	0.0103	0.0947
public 5000, classical cavity	plain	extrap.	0.3250	0.4007	0.0542	0.3167
		interp.	0.2937	0.3961	0.0555	0.3251
public 5000, classical cavity	prior	extrap.	0.4078	0.4898	0.0715	0.3222
		interp.	0.3071	0.3643	0.0532	0.2487

Applying the scaling also changes the failure mode. The plain model's error at an unseen contrast grows with the distance from the training contrast, because its correction is a fixed field. The rescaled model's error stays proportional to the error in P, roughly uniformly in ε.

The paired scan makes the information limit explicit. Within one configuration X is identical for all five contrasts, so an ε-blind model must emit one field for all five. At $\varepsilon = 1.9$ the error is 0.388/0.575/0.060/0.424 for $\hat{Y} = X$, 0.243/0.295/0.031/0.273

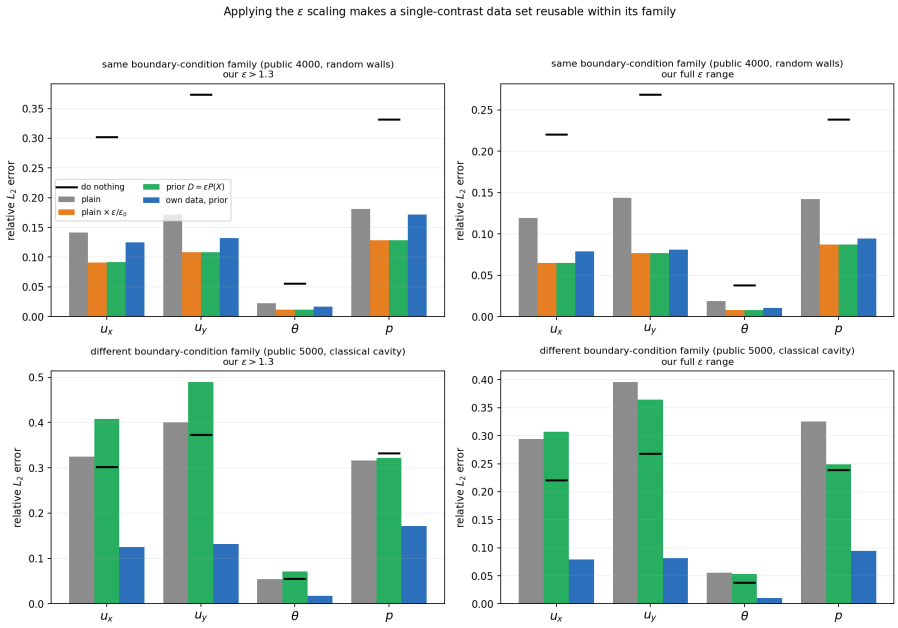


Figure 3: Reusing a single-ε data set. Within the boundary-condition family the prior transfers and beats the unscaled plain model; a plain model with the same scaling applied outside agrees with the factorized target to 0.2% relative (0.0912 vs 0.0914; Table 3). Across families the prior fails and can be worse than returning X. The failure is the flow dependence of A, not the ε scaling. Public-data models are applied with the source per-channel normalisation.

for a model trained only at $\varepsilon = 1.2$, and 0.209/0.268/0.028/0.235 for the unweighted ε-mean field ($u_x/u_y/\theta/p$, median over 20 configurations). No function of X alone can beat the best single field. The missing input is the scalar ε, not network capacity.

Across boundary-condition families the picture is different. Trained on the 5000 classical cavity pairs, the plain model lands at 0.3250 in u_x, worse than the floor of 0.3018, so it learns little that transfers. The prior is worse in extrapolation (0.4078/0.4898/0.0715/0.3222) and does not beat returning X in interpolation on any channel (u_x, u_y, θ or p; 0.3071/0.3643/0.0532/0.2487). This is the predicted boundary: the prior carries the ε dependence, not another family's prefactor. The classical cavity's peak Boussinesq speed is 41% above the random-wall median, so $\hat{u}$ separates the families; it does not order the per-sample error within a family (Spearman −0.20). It is a family-level diagnostic, not a within-family transfer law.

The reverse direction is favorable for the field. The 4000 random-wall set contains the one-mode classical cavity as a special case. The broad-family prior reaches 0.09% relative L_2 on the 200 classical-cavity samples at $\varepsilon = 1.2$, and its wall-flux variant 0.10% (Table 2); the field-only prior's both-wall Nu error still spans 1.4–2.9% across seeds (2.0% seed mean). The asymmetry is coverage, not a violation of the law: a model trained on a narrow family cannot cover a broader one,

while a model trained on the released broad family covers its one-mode subfamily. This is a training-set design rule supported by these families, not a guarantee for an unseen family.

The accuracy trade-off is visible when ε is available at training time. On our own data, the ε-input network is more accurate in extrapolation than the prior; removing ε from the input costs accuracy but buys the ability to apply the model at another contrast. The gap is the price of reuse. At small sample sizes the structured model is also the cheaper one: at $n = 50$ the prior is 1.4× better than the ε-input network in u_x (0.1308 against 0.1831), and it remains ahead at $n = 200$; the crossover is near $n \approx 300$ for u_x and u_y, and θ never crosses. The prior at $n = 200$ reaches what the ε-input network needs about 270–290 samples to reach.

4.3. Wall heat transfer: per-wall breakdown and Rayleigh dependence

Table 2 splits the integral error by wall and adds the local profile error. The hot-wall integral change between the two members is 0.67% (cold wall 0.64%, both walls 0.48%; mean per-sample hot-wall ratio 0.54%, median 0.58%), while the local wall flux is redistributed by 4.25% in L_1 (p90 7.33%, pointwise excursions ±10%). The 4.25% is the L_1 norm of the relative local change $\Delta Nu(y)/Nu(y)$ between X and Y. The 5.6% entry in Table 2 (6.0/5.2% on the two walls) is the mean absolute error of the predicted wall-flux profile divided by its mean absolute value; the two use different denominators and are not directly comparable. The both-wall 2.0% error is the flux-weighted combination of the per-wall 2.0% (hot) and 2.1% (cold) values.

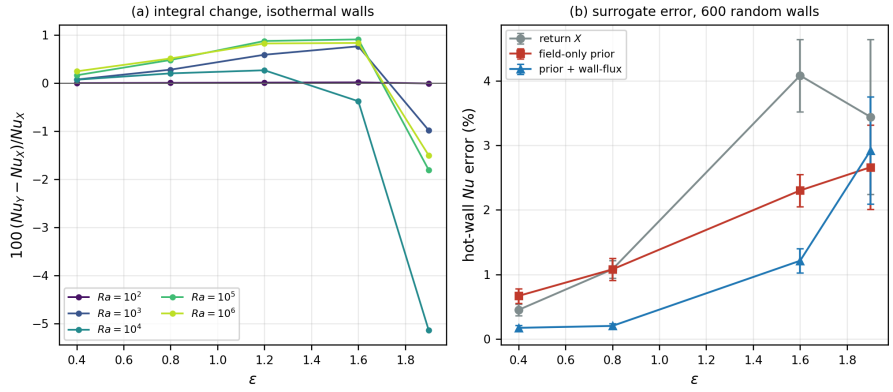


Figure 4: (a) Integral-Nu change between the two members for the five isothermal-wall configurations at $Ra = 10^2$–10^6. The change is not a single function of ε: it peaks at −5.1% at $Ra = 10^4$ and is −1.5 to −1.8% at higher Ra. (b) Hot-wall Nu error of the surrogate on the 600 random-wall samples. The wall-flux penalty reduces the error at intermediate contrast; at $\varepsilon = 1.9$ the improvement is not resolved.

All heat-transfer errors in this section are evaluated on the 64^2 compressible field. The grid audit shows that the pair correction itself changes by 0.43–0.60 percentage points under refinement, so a sub-percent value below measures agreement with

this target, not converged physical accuracy. The public prior reduces the median global temperature-field error from 8.4% to 0.09%, but its both-wall integral Nu error is 2.0% (1.4–2.9% across seeds), larger than the 0.48% of returning X. The wall-normal derivative carries a bias that the field norm does not see. Adding the wall-flux penalty brings the six-point integral error to 0.01% and the local flux error to 0.08%, at a cost of 0.01 percentage point in the median field error (0.09% to 0.10%). When the six-point-trained models are instead evaluated with four- and eight-point stencils, their seed ranges are 0.39–0.49% and 0.46–0.70%, against 1.45% and 2.88% for the field-only prior and 0.39% and 0.90% for returning X. Dedicated training at those two stencils reaches 0.131% and 0.117% (Table 7). At ten points the six-point-trained wall-flux models give 1.00% and 2.21% against the 1.11% floor; seed 0 is worse than returning X. We therefore report no ten-point gain. The ordering is stable at four and eight points, while the absolute value depends on the stencil. At $\varepsilon = 0.4/0.8/1.6$ the penalty gives 0.17/0.20/1.21% against 0.45/1.08/4.08% for returning X; at $\varepsilon = 1.9$ the gain is not resolved.

Table 4 reports the seed-level uncertainty. On the 200 classical-cavity samples, the field-only prior has a both-wall Nu error of 2.04% (95% CI 1.00–3.08%), while the wall-flux model has 0.011% (0.009–0.013%). The paired across-sample difference between the two errors is 2.03 percentage points (1.97–2.10). On the 22 random-wall samples at $\varepsilon = 1.1$–1.3, the corresponding Nu values are 4.04% (2.68–5.40%) and 0.58% (0.45–0.71%). These intervals quantify seed variability; they do not remove the target-grid and matched-stencil limitations.

Table 4: Seed-level means and 95% confidence intervals for the wall-heat-transfer audits. Errors are percentages; the interval is a t interval over three seeds for the classical cavity and five seeds for the random-wall subset.

split	model	temperature field	both-wall Nu	local flux
200 classical	field-only prior	0.094 (0.091–0.096)	2.04 (1.00–3.08)	4.46 (3.69–5.24)
200 classical	prior + wall flux	0.123 (0.119–0.127)	0.011 (0.009–0.013)	0.081 (0.080–0.081)
22 random-wall	field-only prior	0.467 (0.442–0.491)	4.04 (2.68–5.40)	4.94 (3.64–6.25)
22 random-wall	prior + wall flux	0.489 (0.463–0.515)	0.58 (0.45–0.71)	1.14 (1.05–1.23)

The Rayleigh dependence is equally important. On the five isothermal-sidewall configurations, the integral change at $\varepsilon = 1.2$ spans +0.01 to +0.87%. At $\varepsilon = 1.9$ the same five configurations give −0.01, −0.98, −5.14, −1.81 and −1.50% for $Ra = 10^2$, 10^3, 10^4, 10^5 and 10^6 (64^2 values): the change peaks at $Ra = 10^4$ and is −1.5 to −1.8% at higher Ra. A model calibrated on Nu alone does not transfer across Ra; the local field carries the correction. At $Ra = 10^6$ the 256^2 Boussinesq solution gives $Nu = 8.793$ on the hot wall, 0.36% below the benchmark 8.825 of Le Quéré [28]; the 64^2 and 128^2 values are 8.563 and 8.759.

Grid refinement changes the size of the compressible correction (Table 5). For the $c16$ configuration at $Ra = 10^6$, the hot-wall ΔNu at $\varepsilon = 1.2$ is +0.82% at 64^2, +0.56% at 128^2 and +0.44% at 256^2, with a Richardson limit of +0.40%. At $\varepsilon = 1.9$ the same sequence gives −1.50%, −1.88% and −2.10% at 64^2, 128^2 and 192^2. The

64^2 target therefore overstates the $\varepsilon = 1.2$ change by about 0.4 percentage points and understates the $\varepsilon = 1.9$ change by about 0.6 points. The scaling exponent is unaffected: the $64^2/128^2$ ratio of R stays within 1% up to $\varepsilon = 1.2$ and within 3.4% at $\varepsilon = 1.9$ across $Ra = 10^4$–10^6.

Table 5: Grid refinement of the hot-wall $\Delta Nu = (Nu_{\mathrm{comp}} - Nu_{\mathrm{B}})/Nu_{\mathrm{B}}$ for the $c16$ isothermal-sidewall configuration at $Ra = 10^6$. The 256^2 pair at $\varepsilon = 1.9$ did not converge, and the 192^2 run was made only at $\varepsilon = 1.9$. The 64^2 column is the target against which all surrogate errors in this paper are measured.

ε	64^2	128^2	192^2	256^2
1.2	+0.82%	+0.56%	—	+0.44%
1.9	−1.50%	−1.88%	−2.10%	—

4.4. Wall-stencil matrix and three-dimensional consistency checks

The six-point result is not an independent validation because the same stencil defines the training penalty. To separate the metric effect from the estimator, we fit a scaling-only field map $D/\varepsilon = \alpha_c X_c$ from the paired fields and never use a wall gradient in training. We then evaluate integral Nu with four-, six-, eight- and ten-point one-sided stencils. Table 6 shows that this field map is worse than returning X for both-wall Nu at every stencil and on both splits. The result is the intended warning: a correction can be built without wall information and still fail heat transfer.

We also trained the wall-flux U-Net at 4, 6, 8 and 10 points, with two seeds for each target, and evaluated every checkpoint at all four stencils. The NumPy re-execution reproduces the stored four-, six- and eight-point values to within 7.1×10^{-6} absolute error. Table 7 gives the complete two-seed matrix. The minimum in each row is the diagonal cell: training and evaluating on 4, 6, 8 and 10 points gives 0.131%, 0.107%, 0.117% and 0.112%, respectively. The off-diagonal errors are generally at or above the 0.39–1.11% floor of returning X. In particular, the six-point-trained model gives 0.995–2.209% at ten points, while the ten-point-trained model gives 0.657–0.883% at six points. The apparent wall-flux gain is therefore tied to the training estimator rather than to a stencil-independent physical Nu.

The existing three-dimensional ROM provides a second check on field prediction rather than only the scaling exponent. It trains on 12 of the 15 available 3D fields and tests on three held-out flows for each of five contrast values. Table 8 reports the leave-one-contrast results. The structured 3D ROM improves the floor on all four channels, but the velocity-y gain is only 1.26× and the test covers one resolution. It is a field check, not a three-dimensional wall-flux result.

4.5. Cost

Table 9 gives the wall-clock cost. A paired solve splits into a Boussinesq member (median 11.7 s) and a compressible member (22.3 s). The surrogate replaces only

Table 6: Stencil-independent audit of a scaling-only field map. Entries are mean both-wall *Nu* errors (%) on held-out samples. The map is fitted from paired fields only; no wall gradient, four-point, six-point, eight-point or ten-point stencil is used in training.

split	model	4 pt	6 pt	8 pt	10 pt
extrapolation	return X	4.96	4.80	4.82	4.88
extrapolation	scaling-only	5.92	5.76	5.77	5.84
interpolation	return X	3.75	3.57	3.56	3.61
interpolation	scaling-only	4.42	4.26	4.24	4.29

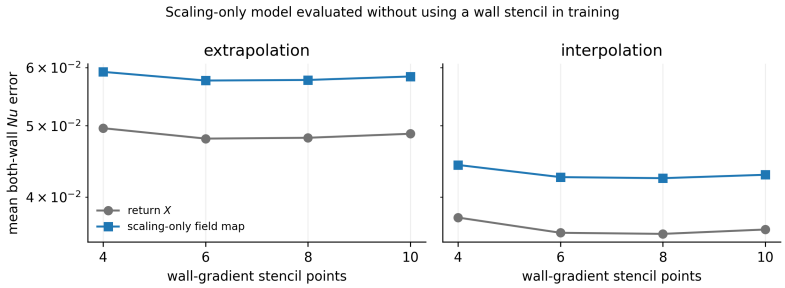


Figure 5: Stencil-independent audit. The scaling-only field map is worse than returning the Boussinesq field for both-wall *Nu* at every evaluation stencil. The ordering is stable; therefore the field-error/Nu-error decoupling is not produced by the six-point estimator.

Table 7: Complete wall-stencil matrix on 200 held-out classical-cavity samples at $\varepsilon = 1.2$. Rows labelled "train" use a wall-flux penalty at that stencil; entries are mean both-wall *Nu* errors (%) over seeds 0 and 1, with exact per-seed values in the supplement. Bold entries are the four diagonal cells and are the row minima. The final row trains against the mean of the four stencil targets; it is not an average of four trained models. All wall-flux models use the same 4000-pair training split.

model	4 pt	6 pt	8 pt	10 pt
return X	0.39	0.67	0.90	1.11
field-only prior	1.45	2.14	2.88	3.93
train at 4 pt	**0.13**	1.26	2.37	3.00
train at 6 pt	0.44	**0.11**	0.58	1.60
train at 8 pt	0.84	0.70	**0.12**	1.57
train at 10 pt	0.67	0.77	0.62	**0.11**
mean-target loss	0.95	0.69	0.27	1.77

Table 8: Leave-one-contrast 3D ROM check. For each of five contrasts, 12 fields train the model and three held-out flows test it. Values are mean relative L_2 errors (± standard deviation across the five leave-one-contrast fits); the last column is the gain over returning the Boussinesq field.

channel	return X	structured 3D ROM	gain
u_x	19.3 ± 13.2%	8.1 ± 6.1%	2.40×
u_y	18.8 ± 12.7%	15.0 ± 10.0%	1.26×
θ	3.62 ± 2.51%	2.70 ± 1.89%	1.34×
p	16.2 ± 11.7%	9.80 ± 7.71%	1.65×

the second. Per pair the cost falls from 33.9 s to 11.7 s, a factor 2.9. For a single flow swept over 100 contrasts, one Boussinesq solve plus 100 inferences costs 16.6 s against 3.77×10^3 for 100 compressible solves, a factor 2.3×10^2. For the 20-flow $\times$ 5-contrast scan, the same-reuse comparison gives 4.07×10^3 s against 321 s, a factor 13. That scan's Boussinesq median is 16.0 s, so 20×16.0 s $+100 \times 14$ ms reproduces the 321 s total. Inference takes 6.3 ms on a GPU and 14.0 ms on one CPU thread.

Table 9: Wall-clock cost on the 64^2 mesh. Solver rows use eight CPU threads; the surrogate runs on an RTX 4060 Laptop GPU (batch 1) or one CPU thread. The 100-sample scan rows use that scan's own timings; medians are not additive. In the 20-flow scan the paired median is 40.7 s per point, so the 100-point baseline is 4.07×10^3 s; its Boussinesq median is 16.0 s, consistent with 20×16.0 s $+100 \times 14$ ms = 321 s. The 20-flow row with reuse includes 20 Boussinesq solves and 100 inferences; its baseline repeats both members at all 100 points.

quantity	median	mean	p90	max
paired solve, 600 samples	33.9 s	38.0 s	59.9 s	510 s
Boussinesq member alone	11.7 s	13.1 s	21.7 s	—
compressible member alone	22.3 s	25.0 s	37.3 s	—
Boussinesq member, 100-sample scan	16.0 s	—	—	—
compressible member, 100-sample scan	37.5 s	—	—	—
paired solve, 100-sample ε scan	55.8 s	67.7 s	136 s	202 s
surrogate, GPU, batch 1	6.3 ms	—	—	—
surrogate, GPU, batch 256	0.49 ms/sample	—	—	—
surrogate, CPU, 1 thread	14.0 ms	—	—	—
end-to-end pair (Boussinesq + surrogate)	11.7 s	—	—	—
20-flow × 5-contrast, paired baseline	4.07×10^3 s	—	—	—
same scan, 1 Boussinesq/flow + 100 inferences	321 s	—	—	—

4.6. Why the scaling is not a fitting artefact

Three independent checks separate the scaling from a flexible network that happened to fit the data. Mesh refinement and solver-noise checks are reported in the Supplementary Material; they do not change the exponent or the sign of the correction.

First, a Fourier neural operator with 1.19M parameters shows the same pattern. Against its ε-blind version, u_x falls from 0.2287 to 0.1334 with the scaling and ε multiplied outside, a factor 1.71. It is statistically indistinguishable from an explicit ε input (0.1334 against 0.1342). On u_y and θ the prior is better; only on p does the explicit input win (0.1536 against 0.1734). The scaling is therefore not tied to a particular backbone.

Second, a POD-ROM with at most four modes on the 15 three-dimensional samples at $Ra = 10^4$ (part of the 33 pairs) improves from 0.168 to 0.081 in u_x when it predicts D/ε instead of D; u_y, θ and p improve by 1.27, 1.25 and 1.67. The gain is largest at small contrast and is not uniform.

Third, one Newton step of the low-Mach residual at the Boussinesq state, on the same discretisation, reproduces the measured correction to median 4.7/3.8/7.2% in $u_x/u_y/p$ on 27 samples at $\varepsilon = 0.02$–0.10, with no extrapolation in ε. The extrapolated P predicts held-out measurements at $\varepsilon = 0.1$ and 0.2 to 2.4/1.8/3.8%. This is the linearised operator doing the work, not an extra fit.

5. Discussion

5.1. What the result does and does not claim

The result is a model-selection rule plus a scaling law, not a universal surrogate. The selection rule is the one that matters for heat transfer: relative L_2 field error can rank a model above doing nothing on integral Nu. The scaling law explains why the two can disagree: the correction is largely a wall-flux redistribution that almost cancels in the integral. The leading exponent is consistent with the same value in the one alternate geometry, one alternate fluid and the three public splits tested; this is not evidence that it is geometry- or fluid-independent beyond those cases. The prefactor is a functional of the flow. Across the three 3-D wall types it varies by factors of 2–13, and changing the fluid from $Pr = 0.71$ to 4.38 moves it by 3.3% for the velocity and temperature channels and 6.1% for pressure. A surrogate that takes X as input can learn the prefactor; a lookup table of A cannot.

This also explains the cross-family failure. The law transfers the ε scaling, which is universal within the retained equations, and not the flow-dependent amplitude. For an engineering application the practical condition is therefore: train on the boundary-condition family and Ra range of interest, then reuse the same model across contrasts without new compressible solves. That is a weaker claim than "train anywhere and apply everywhere", and it is the claim the data support.

5.2. Relation to existing methods

To our knowledge, the closest prior work either learns a correction field at a fixed contrast or treats the contrast as an input channel. Neither yields a scaling law for the gap or a way to reuse a single-contrast data set. Residual-based corrector operators solve a linear variational problem for the residual of a trained operator

and add the solution to its prediction [29, 30]. They are accurate but cost one linear solve per query. The present rule is a single multiplication outside the network; the linear solve is used once, offline, to identify P. Physics-informed extrapolation adds sparse observations or residual terms [31], and transfer operator learning addresses conditional shift [32]. Multi-fidelity surrogates combine cheap and expensive data through learned corrections [33, 34, 35]; they still learn the parameter dependence from samples. The difference here is that the parameter dependence is derived and only the spatial shape is learned.

Dimension-informed operator learning imposes exact units equivariance and proves invariance under rescaling [36]. That is a different invariance: it concerns changes of units, not the physical breaking of the Boussinesq similarity by a finite temperature contrast. The present law is specific to the Boussinesq–compressible pair and predicts the form of the correction, not only its dimensional consistency.

5.3. Practical recipe

For a new flow family:

1. solve the Boussinesq member X on the target family;
2. obtain paired data at one accessible contrast ε_0;
3. train N on $P = D/\varepsilon_0$ with the residual form $\hat{Y} = X + \varepsilon N(X)$, or train a plain model at ε_0 and rescale its output by $\varepsilon/\varepsilon_0$;
4. if wall heat transfer matters, add the wall-flux penalty of Eq. (18);
5. apply at other contrasts without new compressible solves, keeping the family and Ra range fixed.

The cost of the single paired data set is paid once. The model then replaces the compressible member at every other contrast. If the target family is outside the training family, the prefactor must be recalibrated; the ε scaling itself does not need to be relearned. The two implementations above agree to 0.2% relative on the extrapolation split, 0.0912 against 0.0914 (Table 3). The reusable element is the external ε factorisation, not the residual parameterisation or a network-free scaling law.

5.4. Scope

The main sweep is two-dimensional and steady, with $\varepsilon < 2$ and $Ma \lesssim 2 \times 10^{-3}$. The three-dimensional evidence has two parts: 33 pairs at one resolution ($res = 16$) provide an exponent consistency check, and the five leave-one-contrast ROM fits test 15 held-out 3D fields. The latter improves the field floor but has a weak u_y gain and contains no 3D wall-flux target. Neither part establishes a three-dimensional design workflow. The public-transfer evidence is likewise limited to two generator families and one pairs release; it establishes a within-family reuse rule, not general cross-family transfer. The pressure channel is quoted in a fixed gauge. The temperature channel has a flow-dependent amplitude and is not part of the common-exponent

claim. All surrogate field and Nu errors are relative to the 64^2 compressible target Y. That target is not grid-converged: at $Ra = 10^6$ the hot-wall ΔNu changes from 0.82% (64^2) to 0.44% (256^2) at $\varepsilon = 1.2$, and from −1.50% to −2.10% at $\varepsilon = 1.9$ (192^2; Table 5). The 0.01% value is therefore an error against the 64^2 target, not an absolute physical accuracy. The 4×4 matrix spans training and evaluation at 4/6/8/10 points with two seeds per trained target; every row minimum is its diagonal cell. At ten points the six-point-trained model is worse than returning X for seed 0 and only slightly better for seed 1, whereas the ten-point-trained models recover the low error. These distinctions prevent the matched-stencil result from being read as independent physical validation.

6. Conclusion

Relative L_2 field error is not a sufficient selection metric for a heat-transfer surrogate. On 200 held-out classical-cavity samples, a model with a 0.09% median temperature-field error missed the both-wall integral Nusselt number by 2.0% (1.4–2.9% across seeds); in 43% of the samples on average (23–82% across seeds) the field error was below 1% and the Nu error above 2%. On the 22 random-wall samples at $\varepsilon = 1.1$–1.3, the corresponding values are 0.47% and 4.04%, with 66% on average across five seeds (50–77%) in that region. Under the six-point training stencil, a wall-flux penalty reduced the both-wall error to 0.01% and 0.58%, respectively, and the local wall-flux error to 0.08% and 1.14%. The fixed six-point models give 0.39–0.49% and 0.46–0.70% when evaluated at four and eight points; dedicated training at those orders reaches 0.131% and 0.117%. The median field error moved to 0.10% and 0.49%. The six-point values are matched training/evaluation diagnostics, not independent physical accuracy. In the complete two-seed 4×4 matrix, the minimum in every training-stencil row is the matching evaluation stencil: the diagonal means are 0.131, 0.107, 0.117 and 0.112% for 4, 6, 8 and 10 points. The six-point-trained model gives 0.995–2.209% at ten points, while the ten-point-trained model gives 0.657–0.883% at six points. Thus the apparent gain is estimator-specific, not a stencil-independent physical Nu improvement. A scaling-only field map that never uses a wall-gradient penalty is worse than returning X at all four stencils, confirming that field accuracy alone is an unreliable model-selection criterion.

The gap between Boussinesq and compressible steady convection factorises. The relative temperature contrast sets its leading-order amplitude, and the flow sets its spatial shape. Across 627 paired solutions, the velocity channels have nominal exponents close to one, with family-cluster intervals reported in the text; the curvature sign is resolved for velocity and pressure, but its magnitude is not, and the temperature curvature is not resolved. The prefactor is a functional of the flow and changes with geometry and boundary conditions, and weakly with the fluid in the one paired Prandtl-number comparison (3.3% for velocity and temperature, 6.1% for pressure). A 33-sample three-dimensional check at one resolution is

consistent with the same exponent; a separate 15-field leave-one-contrast 3D ROM improves the field floor by 1.26–2.40×, but provides no 3D wall-flux result. Both are consistency checks, not a general three-dimensional validation.

The reusable element is the external ε factorisation. A plain model trained at $\varepsilon_0 = 1.2$ and rescaled by $\varepsilon/\varepsilon_0$ agrees with the factorized target to 0.2% relative on the extrapolation split (0.0912 vs 0.0914); the residual parameterisation is a convenient implementation, not the source of the transfer. The law makes a public single-contrast data set usable within the tested boundary-condition family: the channel-averaged extrapolation error falls from 0.129 to 0.085, 3.1× lower than returning the Boussinesq solution. The gain is not a universal accuracy win over an ε-aware model; it is the ability to apply a model trained at one contrast at another without new compressible solves. The public tests do not establish transfer to an unseen family; the one reverse cross-family experiment fails, as expected from the flow dependence of A.

Accordingly, the learned residual parameterisation should not be presented as a new surrogate architecture. For wall heat transfer, the penalty reduces the matched six-point bias, but its gain can overlap or fall below that of the cheap Boussinesq solution at ten points. All heat-transfer accuracy claims are therefore meaningful at the 0.4–0.6 percentage-point resolution of the 64^2 target, not as absolute wall-flux benchmarks.

Supplementary material

The Supplementary Material contains the five asymptotic checks, the two-parameter (Ra, ε) fits, the exact single-contrast transfer matrix, the learning curve, the public-split statistics, the θ–$\hat{u}$ fits, the three-dimensional fit table, the POD-ROM details, the per-sample field–Nu statistics, the multi-seed transfer and wall-flux tables, mesh-independence and solver-noise checks, and the script-to-result map.

CRediT authorship contribution statement

The authors contributed to conceptualization, methodology, software, validation, formal analysis, investigation, writing (original draft), and writing (review and editing).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used an AI assistant to develop analysis scripts, to draft and edit the manuscript, and to check numerical consistency between the text and the stored results. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

Data availability

The public paired data set of Menglik et al. [27] is available under CC BY 4.0. The curated paired-solution data sets used here (the 627 main pairs and the paired-ε, low-contrast, $Pr = 4.38$ and 2:1-geometry sets), all result files, solver configurations and the analysis scripts that reproduce every number in this paper are deposited at `https://doi.org/10.5281/zenodo.XXXXXXX`, mirrored at `https://github.com/sheep-cloud12138/Algorithms-instead-of-simulations` (branch `dev`); the raw solver output is available from the authors on request.

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